WinterHack 2027 - 3v3 Soccer Agent Strategy Report

Booster Champion simulation | Strategy version: V37 (3.7.0) | 28 October 2026

1. Objective

The aim is to create scoring opportunities more effectively by combining rapid ball recovery, coordinated forward movement and decisive shooting, while preserving a stable defensive structure. The implemented solution addresses three official strategy concepts: Set Piece Offense, Partitioning Strategy, and Movement and Obstacle Avoidance.

2. Direct kickoff and second-ball opportunities (Set Piece Offense)

At our kickoff, a designated striker uses a high-power kick (power 10) toward a relatively open area inside the far goal mouth, with the aiming direction held stable during execution. The partner moves forward to support a rebound or loose ball instead of waiting in a fixed position. A deep kick can also lead to an opponent clearance or restart; after the referee permits play, our forwards attempt to recover the second ball and attack quickly. The current implementation does not deliberately force the ball out of play: that restart sequence is an opportunistic scenario, not a guaranteed outcome.

3. Two-striker coordination (Partitioning Strategy)

The primary attacker is selected dynamically using estimated ball-control arrival time. The other striker provides a separate receiving lane, rather than converging on the same ball. During a counterattack or when an opponent is close to possession, both strikers may press; otherwise one attacks and the other supports. A short receiver-lock window helps avoid competing kick commands and missed passes. This is intended to improve recovery speed, passing options and follow-up shots.

4. Fast finishing and directional control

The agent checks candidate shooting paths through the actual goal width, with clearance and angular-error margins. When a suitable opening exists in attacking territory, it selects a power-8 shot, rather than delaying for unnecessary dribbling or allowing an opponent to close the shooting lane. If a reliable finish is unavailable, it can choose a forward goal-line drive (power 8), a pass (power 1.5 or 4), or a controlled advance. The kick type is chosen according to the situation, not treated as a continuously variable power scale.

5. Goalkeeper interception and safe movement

The goalkeeper tracks the line from the ball to the centre of our goal and estimates where an incoming ball will cross the defensive line. It moves to block that corridor early, presses a nearby loose ball only within a limited safe area, and clears forward when contact is feasible. V37 prioritises turn-and-forward motion over ineffective lateral-only commands and applies referee-aware separation and field-boundary constraints during opponent restarts.

6. Evaluation and evidence

Code validation: 196 offline unit/scenario tests passed. Decision-level comparison: replaying a recorded V36 throw-in state changed a risky approximately 0.63 m ball separation into a planned 2.40 m conservative retreat target. For one historical incoming shot, V37 selected an earlier goalkeeper lateral target (approximately -0.41 m versus +0.04 m in V36). These are replayed decisions, not measured saves. Live log: 11,555 V37 frames were supplied; the observed score progressed from 1-1 to 3-3 (two goals for each team during this interval), with no own-player penalties recorded. These logs demonstrate functioning tactics and recorded scoring, but do not establish a statistically controlled improvement over the unmodified official baseline. A like-for-like match comparison is still needed to quantify that improvement.